

Graph Theory

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1 Trees

Theorem 1

Let T be a graph of order n (vertices). Then the following statements are equivalent:

1. T is a tree.
2. T contains no cycles, and has $n - 1$ edges.
3. T is connected, and has $n - 1$ edges.
4. T is connected, and each edge is a bridge.
5. Any two vertices of T are connected by exactly one path.
6. T contains no cycles, but the addition of any new edge creates exactly one cycle.

Proof. For $n = 1$, all the statements are trivial, so assume $n \geq 2$.

$1 \Rightarrow 2$: Since T contains no cycles, the removal of any edge must disconnect T into two graphs, each of which is a tree. It follows by induction that the number of edges in each of these two trees is one fewer than its number of vertices. We deduce that the total number of edges is

$$(n_1 - 1) + (n_2 - 1) + 1 = n - 1.$$

$2 \Rightarrow 3$: By contradiction: if T is disconnected, then each component of T is a connected graph with no cycles, and hence the number of vertices in each component exceeds the number of edges in that component by exactly 1. Summing over all components, the total number of vertices of T exceeds the total number of edges by at least 2 (since T has at least two components), contradicting the fact that T has $n - 1$ edges.

$3 \Rightarrow 4$: The removal of any edge results in a graph with n vertices and $n - 2$ edges, which must be disconnected. Indeed, if a graph has k components, n vertices, and m edges, then

$$n - k \leq m \leq \frac{(n - k)(n - k + 1)}{2},$$

so for $m = n - 1$ this holds only for $k = 1$.

$4 \Rightarrow 5$: Since T is connected, each pair of vertices is connected by at least one path. If a given pair of vertices were connected by two paths, then they would enclose a cycle, contradicting the fact that each edge is a bridge.

$5 \Rightarrow 6$: If T contained a cycle, then any two vertices on that cycle would be connected by at least two paths, contradicting statement (5). If an edge $e = uv$ is added to T , a cycle is created, since u and v were already connected. Suppose now that adding $e = uv$ created more than one cycle; then there would be two paths from u to v in the original graph T , which contradicts (5).

$6 \Rightarrow 1$: Suppose that T is disconnected. If we add an edge joining a vertex from one component to a vertex of another component, then no cycle is created, a contradiction. $\square$

Theorem 2

If $n \geq 2$, any tree on n vertices has at least two end-vertices (leaves).

Proof. By the Handshaking Lemma, $\sum d_i = 2|E| = 2n - 2$. If every vertex had degree at least 2, we would have $\sum d_i \geq 2n$; since $2n - (2n - 2) = 2$, this shows that there must be at least two vertices of degree 1. $\square$

Theorem 3

If T is any spanning tree of a connected graph G , then:

1. Each cutset of G has an edge in common with T .
2. Each cycle of G has an edge in common with the complement of T .

Proof. (1): Let C^* be a cutset of G , the removal of which splits a component of G into two subgraphs H and K . Since T is a spanning tree, T must contain an edge joining a vertex of H to a vertex of K ; this is the required edge.

(2): Let C be a cycle of G having no edge in common with the complement of T . Then C is contained in T , which is the required contradiction, since T is acyclic. $\square$

Corollary 1 (Spanning Trees in Complete Graphs)

The number of spanning trees of K_n is n^{n-2} .

Remark. No proof was included in the original notes for this result. The notes give only the following intuition: every distinct tree with n vertices can form a spanning tree in a complete graph with n vertices.

2 Connectivity in Graphs

Theorem 1

A graph G is bipartite $\iff$ every cycle of G has even length.

Proof. ($\Rightarrow$) Since G is bipartite, we can split its vertex set into two disjoint sets A and B so that each edge of G joins a vertex of A and a vertex of B . Let $v_0 \rightarrow v_1 \rightarrow \dots \rightarrow v_n \rightarrow v_0$ be a cycle of G , and assume $v_0 \in A$; then $v_1 \in B$, $v_2 \in A$, and so on. Since v_n must lie in B , the cycle has even length.

($\Leftarrow$) Pick any vertex $u \in V(G)$, and define two sets

$$X = \{x \in V : d(u, x) \text{ is even}\}, \quad Y = \{y \in V : d(u, y) \text{ is odd}\},$$

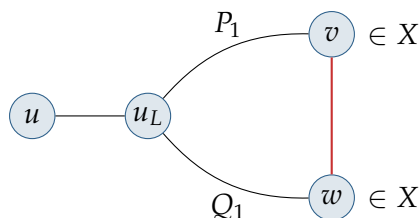
where $d(u, x)$ denotes the length of a shortest path from u to x . We claim (X, Y) is a bipartition, i.e. no edge connects two vertices of X or two vertices of Y .

Suppose, for contradiction, that there exists an edge connecting two vertices in the same set, say $v, w \in X$. Let P be a shortest path from u to v , and Q a shortest path from u to w . Denote by u_L the last common vertex of P and Q (where the two paths diverge), so that

$$P = u \rightarrow \dots \rightarrow u_L \rightarrow P_1 \rightarrow v, \quad Q = u \rightarrow \dots \rightarrow u_L \rightarrow Q_1 \rightarrow w.$$

Both P and Q have the same length up to u_L ; since the lengths of P and Q are both even, the lengths of P_1 and Q_1 are also even (because each total is even). Now consider the path $P_1 + \overline{Q_1} + (v, w)$ (i.e. P_1 , followed by Q_1 reversed, followed by the edge (v, w)): this is a cycle of odd length, a contradiction.

The same argument applies to the set Y .



□

Theorem 2

Let G be a simple graph on n vertices. If G has k components, then the number m of edges of G satisfies

$$n - k \leq m \leq \frac{(n - k)(n - k + 1)}{2}.$$

Proof. We prove the lower bound $m \geq n - k$ by induction on the number of edges; the result is trivial if G is a null graph. If G contains as few edges as possible (say m_0) subject to having n vertices and k components, then the removal of any edge of G must increase the number of components by 1, and the remaining graph has n vertices, $k + 1$ components, and $m_0 - 1$ edges. By the induction hypothesis,

$$m_0 - 1 \geq n - (k + 1) \implies m_0 \geq n - k,$$

as required.

To prove the upper bound, we assume that each component of G is complete. Suppose there are two components C_i, C_j with n_i and n_j vertices respectively, $n_i \geq n_j \geq 1$. If we replace C_i and C_j by complete graphs on $n_i + 1$ and $n_j - 1$ vertices, then the total number of edges

$$\frac{(n_i + 1)(n_i)}{2} + \frac{(n_j - 1)(n_j - 2)}{2}$$

is increased by $n_i - n_j + 1 > 0$. It follows that, in order to maximize the number of edges, G must consist of a complete graph on $n - k + 1$ vertices together with $k - 1$ isolated vertices, which gives

$$\frac{(n - k + 1)(n - k)}{2} + 0 \times (k - 1).$$

□

Corollary 3

Any simple graph with n vertices and more than $\frac{(n - 1)(n - 2)}{2}$ edges is connected.

Proof. By Theorem 2,

$$\frac{(n - 1)(n - 2)}{2} \geq \frac{(n - k)(n - k + 1)}{2}, \quad \forall k \geq 2,$$

which implies $k = 1$.

□

3 Eulerian Graphs and Digraphs

Lemma 1

If G is a graph in which the degree of each vertex is at least two, then G contains a cycle.

Proof 1. If G is a simple graph, let v_0 be any vertex of G . We construct a walk $v_0 \rightarrow v_1 \rightarrow \dots$ inductively, by choosing v_{i+1} to be any vertex adjacent to v_i ; the existence of this vertex is guaranteed by our hypothesis. Since G has only finitely many vertices, we must eventually choose a vertex that has been chosen before, and the walk between its two occurrences is a cycle. $\square$

Proof 2 (sketch). G is not a tree. $\square$

Remark. The notes leave this alternative argument as a brief hint, without further elaboration.

Theorem 2 (Euler, 1736)

A connected graph is Eulerian $\iff$ the degree of each vertex is even.

Proof. ($\Rightarrow$) Suppose that P is an Eulerian trail in G . Whenever P passes through a vertex, there is a contribution of 2 towards the degree of that vertex ($\rightarrow v \rightarrow$); since each edge occurs exactly once in P , each vertex must have even degree.

($\Leftarrow$) The proof is by induction on the number of edges of G . Suppose that the degree of each vertex is even. Since G is connected, each vertex has degree at least 2, and so by Lemma 1, G contains a cycle C . If C contains every edge of G , the proof is complete. If not, we remove from G the edges of C , to form a possibly disconnected graph H with fewer edges than G , in which each vertex still has even degree. By induction, each component of H has an Eulerian trail. Since each component of H has at least one vertex in common with C , by connectedness we obtain the required Eulerian trail of G by following the edges of C until a non-isolated vertex of H is reached, tracing the Eulerian trail of the component of H that contains that vertex, and then continuing along the edges of C ; the whole process terminates when we reach the initial vertex. $\square$

This proof can easily be modified to prove the following results.

Corollary 1

A connected graph is Eulerian $\iff$ it can be split (as a set of edges) into disjoint cycles.

Remark. No separate proof was given in the original notes; the modification of the proof of Theorem 2 is indicated but not written out.

Corollary 2

A connected graph is semi-Eulerian $\iff$ it has exactly two vertices of odd degree.

Proof. ($\Rightarrow$) Assume we have a trail T from u to v . If we add an edge between v and u , we obtain an Eulerian graph (all vertices have even degree); if we remove that edge, only u and v have odd degree.

($\Leftarrow$) We add an edge between these two vertices; then all vertices have even degree, so it is an Eulerian graph. $\square$

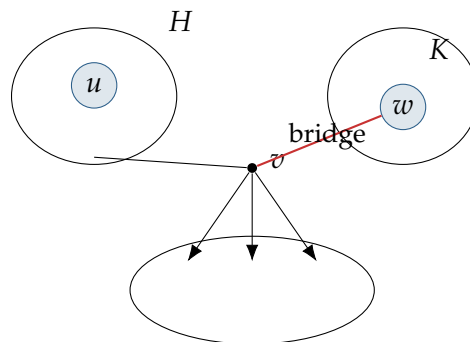
Theorem 3 (Fleury's Algorithm)

Let G be an Eulerian graph. Then the following construction is always possible and produces an Eulerian trail of G . Start at any vertex u and traverse the edges in any way, subject to this rule:

1. Erase each edge as it is traversed, and if any isolated vertex results, erase it too.
2. At each stage, use a bridge only if there exists no alternative.

Proof. We first show that the construction can be carried out at each stage. Suppose that we have just reached a vertex v . If $v \neq u$, then the subgraph H that remains is connected and contains only two vertices of odd degree, u and v . To show that the construction can be carried out, we must show that the removal of the next edge doesn't disconnect H , or equivalently, that v is incident with at most one bridge.

But if this is not the case, then there exists a bridge vw such that the component K of $H - vw$ containing w does not contain u . Since the vertex w has odd degree in K , some other vertex of K must also have odd degree, giving the required contradiction.



If $v = u$, the proof is almost identical, as long as there are still edges incident with u .

It remains only to show that this construction always yields a complete Eulerian trail. But this is clear, since there are no edges of G remaining untraversed when the last edge incident to u is used: otherwise, the removal of some smaller edge adjacent to one of these edges would have disconnected the graph, contradicting rule 2. $\square$

Theorem 4

A strongly connected digraph D is Eulerian $\iff$ for each vertex v of D , $\text{outdeg}(v) = \text{indeg}(v)$.

Proof. ($\Rightarrow$) Suppose D has an Eulerian trail. Each time the trail enters a vertex v , it uses one incoming arc; each time it leaves v , it uses one outgoing arc. Since every arc appears exactly once in the trail, $\text{outdeg}(v) = \text{indeg}(v)$.

($\Leftarrow$) We proceed by induction on the number of arcs; the result is trivial for no arcs. Assume the theorem holds for all connected digraphs with fewer arcs than D . Since D is connected and every vertex satisfies $\text{outdeg}(v) = \text{indeg}(v)$, every vertex has at least one outgoing arc. Starting from any vertex and repeatedly following outgoing arcs, because the graph is finite, we must eventually repeat a vertex. Hence D contains a directed cycle C .

If C contains all arcs, the proof is complete. Otherwise, we remove the arcs of C from D , obtaining a digraph, possibly disconnected, H . Each component of H has fewer arcs than D , so it has an Eulerian trail. Moreover, since D is connected, every component of H shares at least one vertex with the cycle C .

We now construct an Eulerian trail of D : start traversing the directed cycle C ; when a vertex is reached that belongs to H with unused arcs, traverse the Eulerian trail of that component of H , then continue along C ; the resulting trail is a directed Eulerian trail of D . $\square$

4 Hamiltonian Graphs and Digraphs

Theorem 3 (Ore, 1960)

If G is a simple graph with $n \geq 3$ vertices, and $\deg(v) + \deg(w) \geq n$ for every pair of non-adjacent vertices v and w , then G is Hamiltonian.

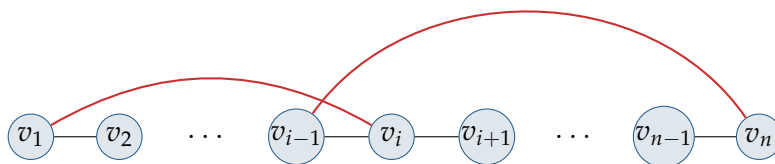
Proof. We assume the statement is false and derive a contradiction. So let G be a non-Hamiltonian graph on n vertices satisfying the given condition on the vertex degrees. By adding extra edges if necessary, we may assume that G is “just barely” non-Hamiltonian, in the sense that adding any further edge produces a Hamiltonian graph (note that adding extra edges cannot violate the degree condition, since degrees can only increase). It follows that G contains a path

$$v_1 \rightarrow v_2 \rightarrow \cdots \rightarrow v_n$$

passing through every vertex. But since G is non-Hamiltonian, the vertices v_1 and v_n are not adjacent, and so $\deg(v_1) + \deg(v_n) \geq n$; by the pigeonhole principle, there must be some vertex v_i adjacent to v_1 with the property that v_{i-1} is adjacent to v_n . But this gives the required contradiction, since

$$v_1 \rightarrow v_2 \rightarrow \cdots \rightarrow v_{i-1} \rightarrow v_n \rightarrow v_{n-1} \rightarrow \cdots \rightarrow v_{i+1} \rightarrow v_i \rightarrow v_1$$

is then a Hamiltonian cycle.



□

Corollary 7.2 (Dirac, 1952)

If G is a simple graph with $n \geq 3$ vertices, and $\deg(v) \geq \frac{n}{2}$ for each vertex v , then G is Hamiltonian.

Proof. The result follows from the preceding theorem, since for all $v, u \in V(G)$ we have $\deg(v) + \deg(u) \geq n$, and hence G is Hamiltonian. □

Theorem (Hamiltonian Digraphs — Ghouila-Houri)

Let D be a strongly connected digraph with n vertices. If $\text{outdeg}(v) \geq \frac{n}{2}$ and $\text{indeg}(v) \geq \frac{n}{2}$ for each vertex v , then D is Hamiltonian.

Remark. No proof was included in the original notes for this theorem.

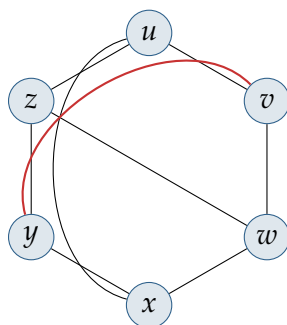
5 Planarity

Theorem 1

The graphs K_5 and $K_{3,3}$ are non-planar.

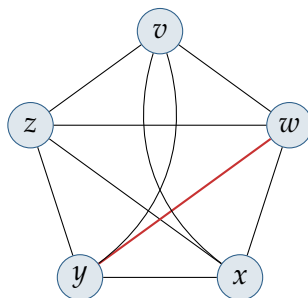
Proof. Suppose first that $K_{3,3}$ is planar. Since $K_{3,3}$ has a cycle $u \rightarrow v \rightarrow w \rightarrow x \rightarrow y \rightarrow z \rightarrow u$ of length 6, any planar drawing must contain this cycle drawn in the form of a hexagon.

Now the edge wz must lie either wholly inside the hexagon or wholly outside it. We deal with the case in which wz lies inside the hexagon — the other case is similar. Since the edge ux must not cross the edge wz , it must lie outside the hexagon. It is then impossible to draw the edge vy , as it would cross either ux or wz .



Now suppose K_5 is planar. Since K_5 has a cycle $v \rightarrow w \rightarrow x \rightarrow y \rightarrow z \rightarrow v$ of length 5, any planar drawing must contain this cycle drawn in the form of a pentagon.

Now the edge wz must lie either wholly inside the pentagon or wholly outside it. We deal with the case in which wz lies inside the pentagon — the other case is similar. Since the edges vx and vy must not cross the edge wz , they must both lie outside the pentagon. But the edge xz cannot cross vy , and so must lie inside the pentagon; similarly, wy cannot cross vx , and so must also lie inside. Then wy and xz must cross, a contradiction.



□

Theorem 2 (Kuratowski, 1930)

A graph is planar if and only if no subgraph is homeomorphic to K_5 or $K_{3,3}$.

Theorem 3

A graph is planar if and only if no subgraph is contractible to K_5 or $K_{3,3}$.

Handshaking Lemma (for Faces)

In any plane drawing of a planar graph,

$$\sum \deg(f_i) = 2 |E(G)|.$$

Proof. In a plane drawing, each edge has two sides (two incident faces):

- the edge may lie on the boundary of a single face (counted twice), or
- the edge lies on the boundary of a face f_1 on one side, and a face f_2 on the other side.

In either case, each edge contributes exactly 2 to the total degree. Hence $\sum \deg(f_i) = 2 |E(G)|$. $\square$

Theorem 4 (Euler, 1750)

Let G be a plane drawing of a connected planar graph, and let n , m , and F denote respectively the number of vertices, edges, and faces of G . Then

$$n - m + F = 2.$$

Proof. By induction on the number of edges of G . If $m = 0$, then $n = 1$, $F = 1$. The theorem is true in this case.

Now, suppose that the theorem holds for all graphs with at most $m - 1$ edges. Let G be a graph with m edges. If G is a tree, then $m = n - 1$, $F = 1$, so that indeed $n - m + F = 2$ as required.

If G is not a tree, let e be an edge in some cycle of G . Then $G - e$ is a connected plane graph with n vertices, $m - 1$ edges, $F - 1$ faces, so that $n - (m - 1) + (F - 1) = 2$ by the induction hypothesis. Adding back the removed edge gives $n - m + F = 2$, as required. $\square$

Corollary 1

- If G is a connected simple planar graph with n (≥ 3) vertices and m edges, then $m \leq 3n - 6$.
- If, in addition, G has no triangle, then $m \leq 2n - 4$.

Proof. (i): Since each face is bounded by at least three edges, it follows, on counting up the edges around each face, that

$$3F \leq 2m$$

(the factor 2 appears since each edge bounds two faces). We obtain the required bound by replacing $F = 2 - n + m$.

(ii): This follows in a similar way, by replacing $3F \leq 2m$ with $4F \leq 2m$. $\square$

As a consequence of Corollary 1, we obtain a second, shorter proof that K_5 and $K_{3,3}$ are non-planar (Theorem 1).

Proof 2 of Theorem 1 (via Euler's Formula). $K_{3,3}$: $n = 6$, $m = 9$ (no triangles). Since $2n - 4 = 8 < m = 9$, $K_{3,3}$ is non-planar.

K_5 : $n = 5$, $m = 10$. Since $3n - 6 = 9 < m = 10$, K_5 is non-planar. $\square$

Corollary 2

Let G be a simple connected planar graph. Then G contains a vertex of degree 5 or less.

Proof. Suppose, for contradiction, that $d_i \geq 6$ for every vertex. Then

$$\sum d_i \geq 6n \implies 2m \geq 6n \implies m \geq 3n > 3n - 6,$$

contradicting $m \leq 3n - 6$ (Corollary 1(i)).

□

6 Graph Colouring

Theorem 1

A graph G is bipartite $\iff \chi(G) = 2$.

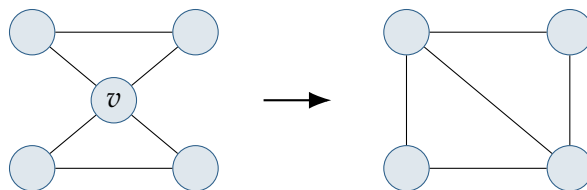
Proof. ($\Rightarrow$) Since G is bipartite, the vertex set can be split into two sets A and B where all edges connect vertices from different sets. Assign colour 1 to the vertices of A and colour 2 to the vertices of B ; this yields a 2-colouring. Clearly a bipartite graph is not 1-colourable (unless it has no edges).

($\Leftarrow$) Since $\chi(G) = 2$, we construct two sets A, B based on colour: put the vertices of colour 1 in set A , and the vertices of colour 2 in set B . Two vertices from the same set are not adjacent, since they have the same colour. Hence G is a bipartite graph. $\square$

Theorem 2

If G is a simple graph with largest vertex degree Δ , then G is $(\Delta + 1)$ -colourable.

Proof. The proof is by induction on the number of vertices of G . Let G be a simple graph with n vertices. If we delete any vertex v and its incident edges, the graph that remains is a simple graph with $n - 1$ vertices and largest vertex degree at most Δ . By the induction hypothesis, this graph is $(\Delta + 1)$ -colourable. A $(\Delta + 1)$ -colouring for G is then obtained by colouring v with a colour different from the vertices adjacent to v .



$\square$

Theorem 3

Let G be a connected graph whose maximum vertex degree is d . If G is neither a cycle graph with an odd number of vertices nor a complete graph, then $\chi(G) \leq d$.

Remark. This is Brooks' Theorem. (No proof was included in the original notes.)

Theorem 4

The vertices of any simple connected planar graph G can be coloured with six (or fewer) colours.

Proof. We prove the theorem by induction on the number of vertices. The result is trivial for simple planar graphs with at most 6 vertices. Suppose then that G is a simple planar graph with n vertices, and that all planar graphs with $n - 1$ vertices are 6-colourable.

We know that G contains a vertex v of degree at most 5; if we delete v and its incident edges, then the graph that remains has $n - 1$ vertices, and is thus 6-colourable. Then a 6-colouring of G is obtained by colouring v with a colour different from the vertices adjacent to v . $\square$

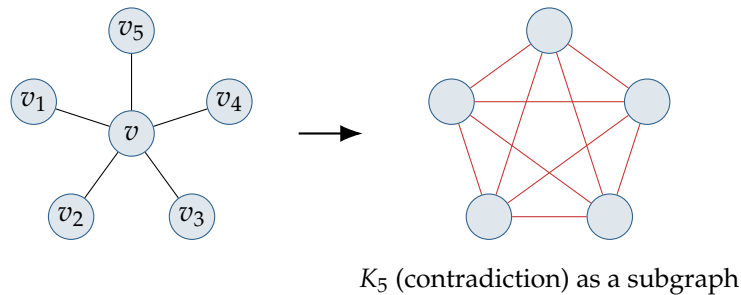
Theorem 5

The vertices of any simple connected planar graph can be coloured with five (or fewer) colours.

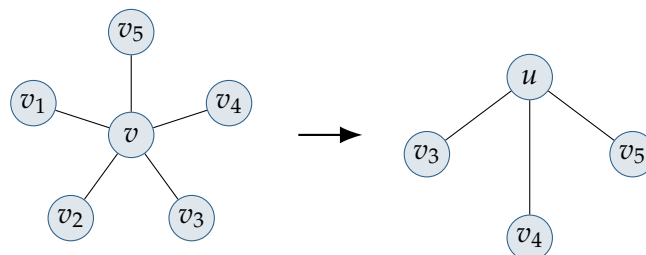
Proof. Induction on the number of vertices. The result is trivial for simple planar graphs with fewer than six vertices, and we assume that all simple planar graphs with $n - 1$ vertices are 5-colourable. G contains a vertex v of degree at most 5. As before, the deletion of v leaves a graph with $n - 1$ vertices, which is thus 5-colourable. Our aim is to colour v with one of the five colours.

If $\deg(v) < 5$, then v can be coloured with any colour not assumed by the (at most four) vertices adjacent to v , completing the proof.

If $\deg(v) = 5$, and the vertices $v_1, \dots, v_5$ adjacent to v are arranged around v in clockwise order: if the vertices are mutually adjacent, then G contains the non-planar graph K_5 as a subgraph, which is impossible.



So at least two of the vertices v_i are not adjacent (say v_1 and v_2). We contract the two edges vv_1 and vv_2 ; the resulting graph is a planar graph with fewer than n vertices, and is thus 5-colourable.



We now reinstate the two edges, giving both v_1 and v_2 the colour originally assigned to v . A 5-colouring of G is then obtained by colouring v with a colour different from the (at most four) colours assigned to the vertices v_i . □

Theorem 6 (Four Colour Theorem)

The vertices of any simple connected planar graph can be coloured with four (or fewer) colours.

Remark. Stated without proof in the original notes (the theorem's known proof is computer-assisted).